

# Chapter 123

# Warp Nodes

This chapter details the Warp nodes available in Fusion.

The abbreviations next to each node name can be used in the Select Tool dialog when searching for tools and in scripting references.

For purposes of this document, node trees showing MediaIn nodes in DaVinci Resolve are interchangeable with Loader nodes in Fusion Studio, unless otherwise noted.

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# Coordinate Space [CdS]



The Coordinate Space node

## Coordinate Space Node Introduction

The Coordinate Space node changes the coordinate space of the image from rectangular to polar or from polar to rectangular.

### Inputs

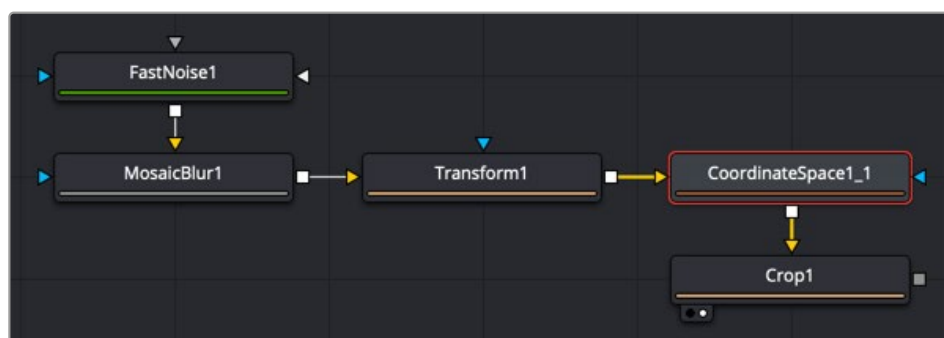
The two inputs on the Coordinate Space node are used to connect a 2D image and an effect mask, which can be used to limit the distorted area.

**Input:** The orange input is used for the primary 2D image that is distorted.

**Effect Mask:** The blue input is for a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the distortion to only those pixels within the mask. An effects mask is applied to the tool after the tool is processed.

### Basic Node Setup

The Coordinate Space node is used below to make a circular pattern based on a Fast Noise, Mosaic Blur (DaVinci Resolve Resolve FX only), and a Transform node. The Crop node at the end is used to set the desired resolution.



The Coordinate Space node can help create motion graphics backgrounds

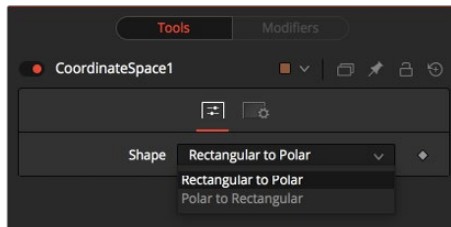
### Example

To demonstrate a basic tunnel effect that can be achieved with this node:

1. Add a Text+ node with some text, and then animate it to move along a path from the top of the frame to the bottom.
2. Connect the output of the Text+ node to a Coordinate Space node.
3. Select Polar to Rectangular from the Shape menu.

As the text moves from top to bottom along the original path, it appears to move from an infinite distance in the Coordinate Space node. It may be necessary to flip the text using a Transform node to make it appear the correct way in the Coordinate Space node. Another common use for the Coordinate Space node is to use it in pairs: two of them set to different Shape settings with a Drip or Transform node in between. When used in this way, the effect gets modified while the image remains the same.

## Inspector



The Coordinate Space Controls tab

### Controls Tab

The Controls tab Shape menu switches between Rectangular to Polar and Polar to Rectangular. Consider the following example to demonstrate the two coordinate spaces.

### Common Controls

#### Settings Tab

The Settings tab in the Inspector is also duplicated in other Warp nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

# Corner Positioner [CPn]



The Corner Positioner node

## Corner Positioner Node Introduction

The Corner Positioner can be used to position the four corners of an image interactively. This would typically be used to replace a sign or other rectangular portion of a scene. Connect all corners to Paths or Trackers for animation purposes.

### Inputs

The two inputs on the Corner Positioner node are used to connect a 2D image and an effect mask, which can be used to limit the warped area.

**Input:** The orange input is used for the primary 2D image that is warped.

**Effect Mask:** The blue input is for a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the Corner Positioner to only those pixels within the mask. An effects mask is applied to the tool after the tool is processed.

### Basic Node Setup

Below, the Corner Positioner is used to position the rectangular corners of the MediaIn2 to fit within a specific area of the MediaIn1 node. A Planar Tracker tracks the background, and then a Planar Transform is used to keep the Corner Positioner in place as the background clip moves. Once the planar tracking is completed, and the Planar Transform is created, the Planar Tracker node is no longer needed, and you can delete it.



The Corner Positioner corner pins a clip in place and uses a Planar Transform to match move it to the background.

## Inspector



The Corner Positioner Controls tab

### Controls Tab

The Controls tab includes transform and offset adjustments for the four corners of the image

#### Mapping Type

This determines the method used to project the image caused by the Corner Positioner. In Bi-Linear mode, a straight 2D warping takes place. In Perspective mode, the image is calculated with the offsets in 2D space and then mapped into a 3D perspective.

#### Corners X and Y

There are four points in the Corner Positioner. Drag these around to position each corner of the image interactively. Attach these control points to any of the usual modifiers.

The image input is deformed and perspective corrected to match the position of the four corners.

#### Offset X and Y

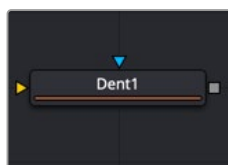
These controls can be used to offset the position of the corners slightly. This is useful when the corners are attached to Trackers with patterns that may not be positioned exactly where they are needed.

### Common Controls

#### Settings Tab

The Settings tab in the Inspector is also duplicated in other Warp nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

# Dent [Dnt]



The Dent node

## Dent Node Introduction

The Dent node creates a circular deformation of an image similar to a Fish Eye Lens effect, with the choice of six different Dent filters.

## Inputs

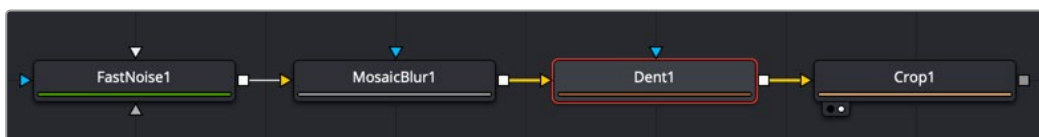
The two inputs on the Dent node are used to connect a 2D image and an effect mask, which can be used to limit the warped area.

**Input:** The orange input is used for the primary 2D image that is warped.

**Effect Mask:** The blue input is for a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the Dent to only those pixels within the mask. An effects mask is applied to the tool after the tool is processed.

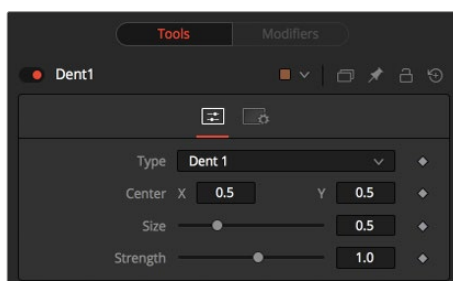
## Basic Node Setup

The Dent node is used below to make a circular pattern based on a Fast Noise and a Mosaic Blur (DaVinci Resolve Resolve FX only). The Crop node at the end is used to set the desired resolution.



The Dent node can help create lens distortion effects or a motion graphics background.

## Inspector



The Dent Controls tab

## Controls Tab

The adjustments in the Controls tab are used to change the Dent style, position, size, and strength.

**Type**

Select the type of Dent filter to use from this menu. All parameters for the Dent can be keyframed.

**Dent 1**

This creates a bulge dent.

**Kaleidoscope**

This creates a dent, mirrors it, and inverts it.

**Dent 2**

This creates a displacement dent.

**Dent 3**

This creates a deform dent.

**Cosine Dent**

This creates a fracture to a center point.

**Sine Dent**

This creates a smooth rounded dent.

**Center X and Y**

This positions the Center of the Dent effect on the image. The default values are 0.5, 0.5, which center the effect in the image.

**Size**

This changes the size of the area affected by the dent. Animate this slider to make the dent grow.

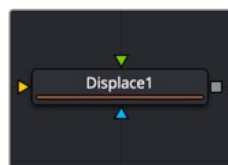
**Strength**

This changes the overall strength of the dent.

**Common Controls****Settings Tab**

The Settings tab in the Inspector is also duplicated in other Warp nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

## Displace [Dsp]



The Displace node

## Displace Node Introduction

The Displace node uses a map image to displace or refract another image. This is useful for creating a vast range of effects from bevels and heat distortion to glass and water effects.

## Inputs

There are three inputs on the Displace node: The primary image, the displacement map foreground image, and an effect mask.

**Input:** The orange image input is a required connection for the primary image you wish to displace.

**Foreground Image:** The green input is also required as the image used to displace the background. Once connected, you can choose red, green, blue, alpha, or luminance channel to create the displacement.

**Effect Mask:** The optional blue effect mask input expects a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the displacement to only those pixels within the mask. An effects mask is applied to the tool after it is processed.

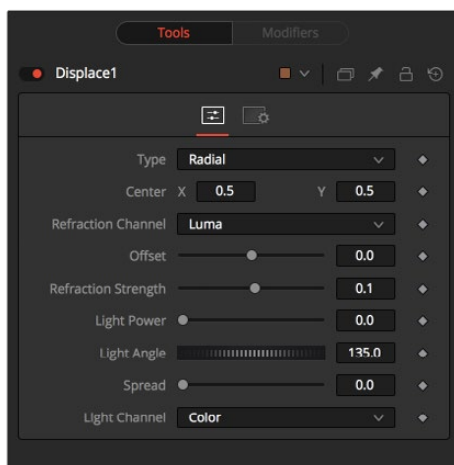
## Basic Node Setup

Below, the Displace node uses a Fast Noise node to generate the Displace map. Increasing the seethe rate can produce heat distortion or flag waving effects.



The Displace node using a Fast Noise node for the Displace map

## Inspector



The Displace Controls tab

### Controls Tab

The Controls tab is used to change the style, position, size, strength, and lighting (embossing) of the displacement.

#### Type

The Type menu is used to choose in what mode the Displace node operates. The Radial mode uses the map image that refracts each pixel out from the center, while X/Y mode provides control over the amount of displacement along each axis individually.



**NOTE:** There is one set of Refraction controls while in Radial mode, and two sets in XY mode—one for each of the X and Y channels.

### Center (Radial Only)

The Center control defines the point from which pixels are displaced toward or away.

### Refraction Channel

This drop-down menu controls which channel from the foreground image is used to displace the image. Select from Red, Green, Blue, Alpha, or Luminance channels. In XY mode, this control appears twice, once for the X displacement and once for the Y displacement.

### Refraction Strength (Radial)

Controls the strength of the refraction. Higher values cause stronger or more pronounced refraction.

### X and Y Refraction (X/Y)

Two separate sliders appear to control the Refraction strength along the X- and Y-axis separately. Otherwise, this is exactly like Refraction Strength.

### Light Power

This controls the intensity, or strength, of the simulated light, causing bright and dim areas to form according to the contour of the refraction image. Higher values cause the bright and dim areas to be more pronounced.

### Light Angle

This sets the angle of the simulated light source.

### Spread

This widens the Displacement effect and takes the edge off the Refraction map. Higher values cause the ridges or edges to spread out.

### Light Channel

Select the channel from the refraction image to use as the simulated light source. Select from Color, Red, Green, Blue, Alpha, or Luminance channels.

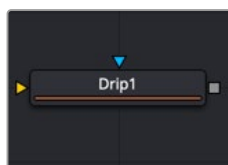
**NOTE:** The Radial mode pushes pixels inward or outward from a center point, based on pixel values from the Displacement map. The XY mode uses two different channels from the map to displace pixels horizontally and vertically, allowing more precise results. Using the XY mode, the Displace node can even accomplish simple morphing effects. The Light controls allow directional highlighting of refracted pixels for simulating a beveled look.

## Common Controls

### Settings Tab

The Settings tab in the Inspector is also duplicated in other Warp nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

# Drip [DRP]



The Drip node

## Drip Node Introduction

The Drip node creates a ripple effect over the entire image, which has the potential to animate outward from a central source. There are a variety of different Drip effects from which to choose.

## Inputs

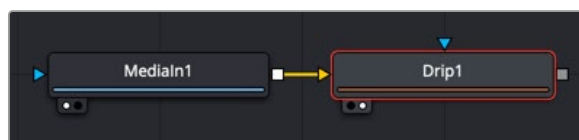
The two inputs on the Drip node are used to connect a 2D image and an effect mask, which can be used to limit the warped area.

**Input:** The orange input is used for the primary 2D image that is warped.

**Effect Mask:** The blue input is for a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the warping to only those pixels within the mask. An effects mask is applied to the tool after the tool is processed.

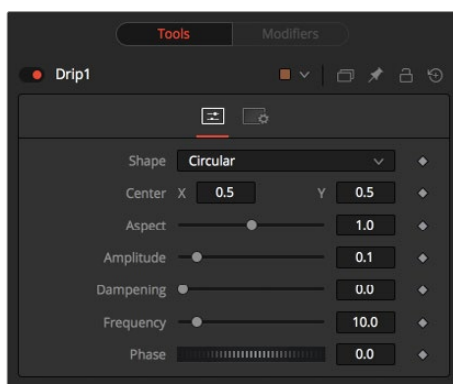
## Basic Node Setup

Below, the Drip node is used to make rippling water-style effects using a MediaIn node.



The Drip node can be connected directly after a MediaIn node or any node providing a 2D output.

## Inspector



The Drip Controls tab

## Controls Tab

The Controls tab is used to change the style, position, size, strength, and phase for animating the “ripples” of the Drip.

**Shape**

Use this control to select the shape of the Drip.

**Circular**

This creates circular ripples. This is the default Drip mode.

**Square**

This creates even-sided quadrilateral drips.

**Random**

This creates a randomly dispersed noise that distorts your image and is similar to a particle effect.

**Horizontal**

This creates horizontal waves that move in one direction.

**Vertical**

This creates vertical waves that move in one direction.

**Exponential**

This creates a Drip effect that looks like a diamond shape with inverted, curved sides (an exponential curve flipped and mirrored).

**Star**

This creates an eight-way symmetrical star-shaped ripple that acts as a kaleidoscope when the phase is animated.

**Radial**

This creates a star-shaped ripple that emits from a fixed pattern.

**Center X and Y**

Use this control to position the center of the Drip effect in the image. The default is 0.5, 0.5, which centers the effect in the image.

**Aspect**

Control the aspect ratio of the various Drip shapes. A value of 1.0 causes the shapes to be symmetrical. Smaller values cause the shape to be taller and narrower, while larger values cause shorter and wider shapes.

**Amplitude**

The Amplitude of the Drip effect refers to the peak height of each ripple. Use the slider to change the amount of distortion the Drip applies to the image. A value of 0.0 gives all ripples no height and therefore makes the effect transparent. A maximum Amplitude of 10 makes each ripple extremely visible and completely distorts the image. Higher numbers can be entered via the text entry boxes.

**Dampening**

Controls the Dampening, or falloff, of the amplitude as it moves away from the center of the effect. It can be used to limit the size or area affected by the Drip.

**Frequency**

This changes the number of ripples emanating from the center of the Drip effect. A value of 0.0 indicates no ripples. Move the slider up to a value of 100 to correspond with the density of desired ripples.

## Phase

This controls the offset of the frequencies from the center. Animate the Phase value to make the ripple emanate from the center of the effect.

## Common Controls

### Settings Tab

The Settings Tab in the Inspector is also duplicated in other Warp nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

# Grid Warp [Grd]



The Grid Warp node

## Grid Warp Node Introduction

The Grid Warp node is a 2D deformation grid with flexible vertices. The image is deformed so that the source grid matches the destination grid.

## Inputs

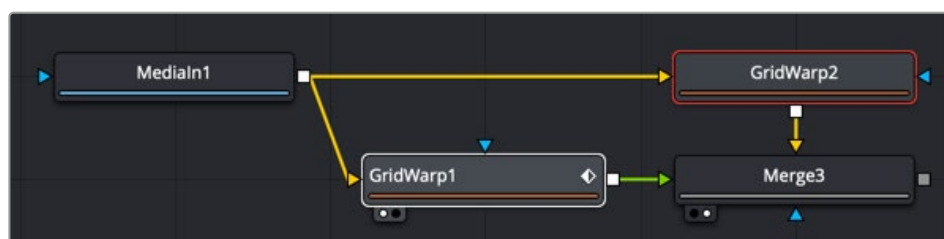
The two inputs on the Grid Warp node are used to connect a 2D image and an effect mask, which can be used to limit the warped area.

**Input:** The orange input is used for the primary 2D image that is warped.

**Effect Mask:** The blue input is for a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the warping to only those pixels within the mask. An effects mask is applied to the tool after the tool is processed.

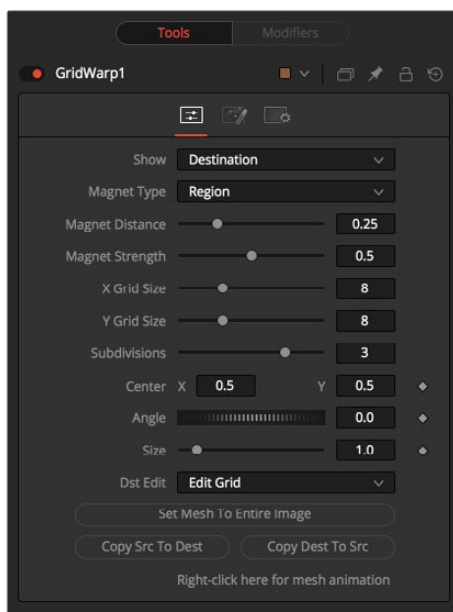
## Basic Node Setup

Below, two Grid Warp nodes are used to warp different areas of a frame. This can be useful for chaining the framing in a shot or adding slight movement to a still. Using the Copy Src to Dest button, only the area modified by Grid Warp 2 is pasted as the foreground in the Merge.



The Grid Warp can be used to shift areas within a shot for reframing or adding animation to a still.

## Inspector



The Grid Warp Controls tab

### Controls Tab

The Controls tab contains parameters that configure the onscreen grid as well the type of distortion applied when a control point on the grid is moved.

#### Source and Destination

The Source and Destination buttons determine whether the source grid or destination grid is currently active. Only one grid can be displayed or manipulated at a time. The selected button is highlighted to indicate that it is the currently active grid.

All other controls in this tab affect the grid selected by this control.

#### Selection Type

These three buttons determine the selection types used for manipulating the points. There are three options available.

#### Selected

When in Selected mode, adjustments to the grid are applied only to the currently selected points. This mode is identical to normal polyline operation.

#### Region

In Region mode, all points within the area around the mouse pointer move when the mouse button is clicked. New points that enter the region during the move are ignored. Choosing this option exposes Magnet Distance and Magnet Strength controls to determine the size and falloff of the area.

#### Magnetic

In Magnetic mode, all points within the area around the mouse pointer move when the mouse button is clicked. New points that enter the region during the move are affected as well. Choosing this option exposes Magnet Distance and Magnet Strength controls to determine the size and falloff of the area.

### Magnet Distance

The default node for selecting and manipulating the grid is a Magnet node. The magnet is represented in the viewer by a circle around the mouse pointer. The Magnet Distance slider controls how large the region of effect for the magnet is, as in the size of the circle. Drag on the grid and any vertex within the range of the slider moves.

To increase the size of the magnet, increase the value of this slider. Alternately, adjust the size of the magnet by holding down the D key while dragging the mouse.

### Magnet Strength

The Magnet Strength slider increases or decreases the falloff of the magnet cursor's effect. At a setting of 0.0, the magnetic cursor has no effect, and vertices do not move at all. As the values increase, the magnet causes a greater range of motion in the selected vertices. Use smaller values for a more sensitive adjustment and larger values for broad-sweeping changes to the grid.

### X and Y Grid Size

The X and Y Grid Size sliders control the number of divisions in the grid. Where the X and Y divisions intersect, a control vertex is created.

Be aware that changing either of these controls after applying changes in the grid resets the entire grid. Set the X and Y grid sizes to the appropriate resolution before making detailed adjustments to the grid.

### Subdivision Level

The Subdivision Level determines how many subdivisions there are between each set of divisions. Subdivisions do not generate vertices at intersections. The more subdivisions, the smoother the deformation is likely to be, but the slower it is to render.

### Center

The Center coordinates determine the exact center of the grid. The onscreen Center control is invisible while editing the grid. Select the Edit Rect mode, and the grid center becomes visible and available for editing.

Use the Center control to move the grid through a scene without affecting the animation applied to the individual vertices. For example, while deforming lips, track the motion of the face with a Tracker, and connect the grid center to the Tracker. This matches the grid with slight movements of the head while focusing on the deformation of the lips.

### Angle

This Angle control rotates the entire grid.

### Size

The Size control increases or decreases the scale of the grid.

### Edit Buttons

There are four edit modes available, each of which can be selected by clicking on the appropriate button.

#### Edit None

Set the grid to Edit None mode to disable the display of all onscreen controls.

#### Edit Grid

The Edit Grid mode is the default mode. While this mode is enabled, the grid is drawn in the viewer, and the control vertices of the grid can be manipulated directly.

### Edit Rectangle

When the grid is in Edit Rectangle mode, the onscreen controls display a rectangle that determines the dimensions of the grid. The sides of the rectangle can be adjusted to increase or decrease the grid's dimension. This mode also reveals the onscreen Center control for the grid.

### Edit Line

The Edit Line mode is beneficial for creating grids around organic shapes. When this mode is enabled, all onscreen controls disappear, and a spline can be drawn around the shape or object to be deformed. While drawing the spline, a grid is automatically created that best represents that object. Additional controls for Tolerance, Over Size, and Snap Distance appear when this mode is enabled. These controls are documented below.

### Set Mesh to Entire Image

The Set Mesh to Entire Image button automatically resets the size of the grid to the exact dimensions of the image. Any adjustments to vertices within the grid are reset.

### Copy Buttons

These two buttons provide a technique for copying the exact shape and dimensions of the source grid to the destination, or the destination grid to the source. This is particularly useful after setting the source grid to ensure that the destination grid's initial state matches the source grid before beginning a deformation.

### Point Tolerance

This control is visible only when the Edit Line mode is enabled. The Point Tolerance slider determines how much tessellation the grid applies to match the density of points in the spline closely. The lower this value, the fewer vertices there are in the resulting grid, and the more uniform the grid appears. Higher values start applying denser grids with variations to account for regions in the spline that require more detail.

### Oversize Amount

This control is visible only when the Edit Line mode is enabled. The Oversize Amount slider is used to set how large an area around the spline should be included in the grid. Higher values create a larger border, which can be useful when blending a deformation back into the source image.

### Snap Distance

This control is visible only when the Edit Line mode is enabled. The Snap Distance slider dictates how strongly the drawn spline attracts surrounding vertices. If a vertex is close enough to a spline's edge, the vertex moves to line up with the spline. The higher the value, the farther the reach of the spline.

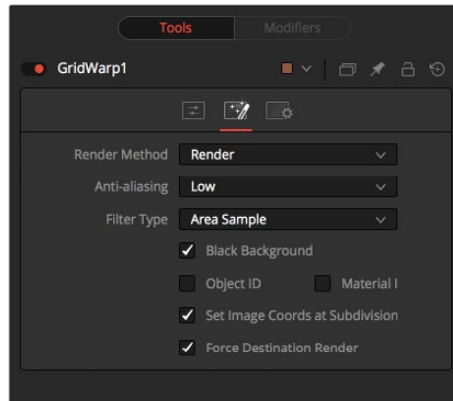
### Right-Click Here for Mesh Animation

The grids are static by default. Right-clicking on the Right-Click Here for Mesh Animation label provides a contextual menu with options for animating the grid or connecting it to another grid in the composition.

The grid uses a Polychange spline. Any adjustment to the control points adds or modifies the keyframe for all points on that spline.

### Right-Click Here for Shape Animation

This label appears only in the Edit Line mode. Right-clicking on the Right-Click Here for Shape Animation label reveals a pop-up menu used to animate the shaping polyline or to connect it to other polylines.



The Grid Warp Render tab

## Render Tab

The Render tab controls the final rendered quality and appearance of the warping.

### Render Method

The Render Method drop-down menu is used to select the rendering technique and quality applied to the mesh. The three settings are arranged in order of quality, with the first, Wireframe, as the fastest and lowest of quality. The default mode is Render, which produces final resolution, full-quality results.

### Anti-Aliasing

The Anti-Aliasing control appears only as a checkbox when in Wireframe Render mode.

In other modes, it is a drop-down menu with three levels of quality. Higher degrees of anti-aliasing improve image quality dramatically but vastly increase render times. The Low setting may be an appropriate option while setting up a large dense grid or previewing a node tree, but rarely for a final render.

### Filter Type

When the Render Method is set to something other than Wireframe mode, the Filter Type menu is visible and set to Area Sample. This setting prevents the grid from calculating area samples for each vertex in the grid, providing good render quality. Super Sample can provide even better results but requires much greater render times.

### Wireframe Width

This slider appears only when the Render Method is set to Wireframe. It determines the width of the lines that make up the wireframe.

### Anti-Aliased

This checkbox appears only when the Render Method is set to Wireframe. Use this checkbox to enable/disable anti-aliasing for the lines that make up the wireframe.

### Black Background

The Black Background checkbox determines whether pixels outside of the grid in the source image are set to black or if they are preserved.

### Object ID and Material ID

Enable the Object ID or Material ID checkboxes to have the grid output the proper ID channel in the final render.

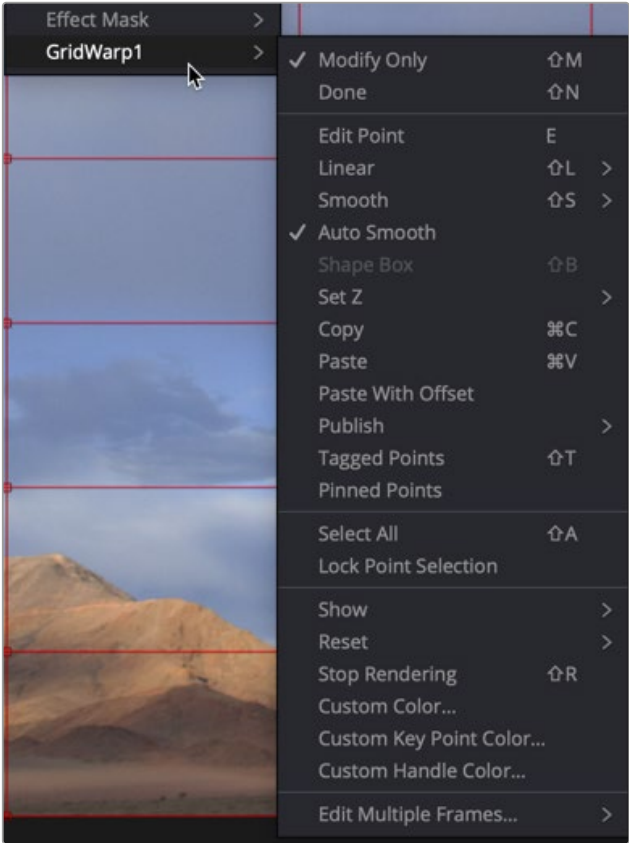


**Set Image Coordinates at Subdivision Level**

This checkbox defaults to enabled and sets the image coordinates at the subdivision level.

**Force Destination Render**

This checkbox defaults to enabled and forces a destination render.



Grid Warp Contextual Menu options

**Contextual Menu Options**

The Grid Warp node places a submenu for both source and destination grids in the viewer’s contextual menu. Both menus have the exact same name, where only the menu for the active grid is populated with options. The other menu is empty. The contextual menu options are also available from the toolbar that appears in the viewer.

**Modify Only/Done**

These two options set the mesh to Modify Only and Done modes, respectively. Select Modify Only to edit the mesh or Modify Done to prevent any further changes to a mesh.

**Smooth/Linear**

Use Smooth and Linear options to apply or remove smoothing from selected vertices.

**Auto Smooth Points**

When Auto Smooth Points is enabled, the vertices in the grid are automatically smoothed whenever they are moved. This is generally enabled by default.

### Z Under/Z Same/Z Over

When two vertices in a grid overlap, one ends up getting clipped by the other. Z Under, Z Same, and Z Over are used to select which vertices are rendered on top and which are rendered behind.

### Select All

This option selects all points in the mesh.

### Show Key Points, Handles, Grid, and Subdivisions

Use these four options to enable or disable the display of the grid, key points (vertices), Bézier handles, and subdivisions in the viewers.

### Reset Selected Points

This resets Selected Points (vertices) to their default positions.

### Reset All Points

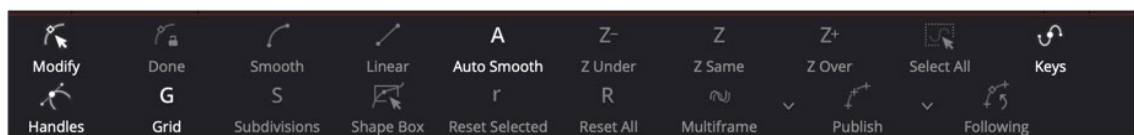
This resets all points (vertices) in the mesh to their default positions.

### Stop Rendering

This option stops rendering, which disables all rendering of the Grid Warp node until the mode is turned off. This is frequently useful when making a series of fine adjustments to a complex grid.

## Grid Warp Toolbar

Whenever the Grid Warp node is selected and is in Edit Grid mode, the Grid Warp toolbar is displayed in the views. This toolbar provides a variety of options for manipulating and adjusting the grid. The toolbar buttons in this toolbar are described in the preceding “Contextual Menu Options” section.



The Grid Warp viewer toolbar

## Common Controls

### Settings Tab

The Settings tab in the Inspector is also duplicated in other Warp nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

# Lens Distort [Lens]



The Lens Distort node

## Lens Distort Node Introduction

This node can be used to remove or add lens distortion in an image. The lens distortion in an image depends on the lens, quality, number of elements, and many other factors.

One reason to remove lens distortion is to composite with an undistorted layer. For example, compositing a 3D element over a distorted live-action layer will cause unwanted effects like straight lines not matching up on the foreground and background. The resulting composite will not look believable.

## Inputs

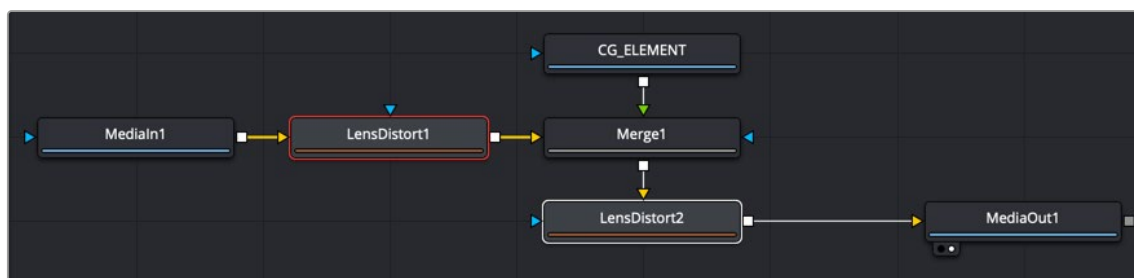
The two inputs on the Lens Distort node are used to connect a 2D image and an effect mask, which can be used to limit the distorted area.

**Input:** The orange input is used for the primary 2D image that is distorted.

**Effect Mask:** The blue input is for a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the distortion to only those pixels within the mask. An effects mask is applied to the tool after the tool is processed.

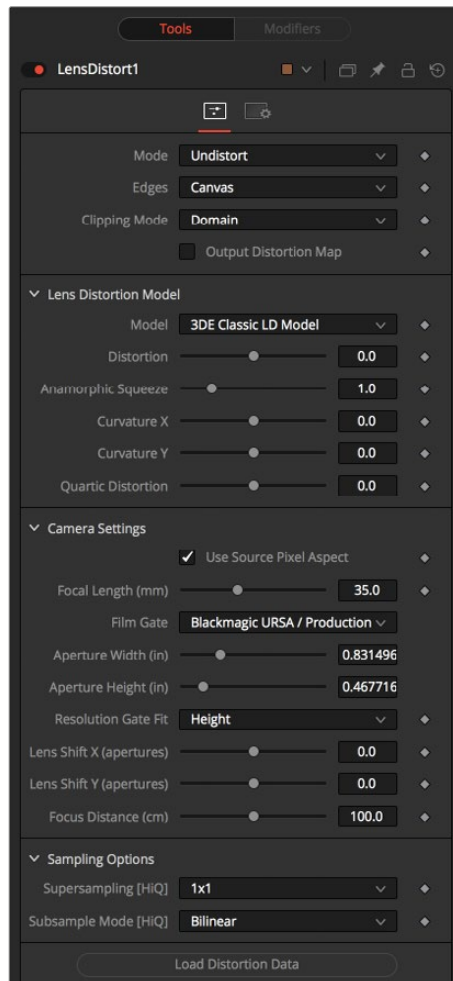
## Basic Node Setup

A simplified example below applies the Lens Distort in Undistort mode to the MediaIn1 live-action layer, composites the 3D elements, and finally applies the Lens Distort at the end with the same settings, but this time in Distort mode to get the original look and distortion back into the image.



Lens Distort applied on the live-action media at the beginning of the node tree, and once again at the end

## Inspector



The Lens Distort Controls tab

### Controls Tab

The Controls tab presents various ways to customize or build the lens distortion model you want. Camera Settings allow you to specify the camera used to capture the content.

#### Mode

Undistort removes the lens distortion to create a flattened image. Distort brings the original lens distortion back into the image.

#### Edges

Determines how samples that fall outside the frame are treated.

- **Canvas:** Pixels outside the frame are set to the default canvas color. In most cases, this is black with no alpha.
- **Duplicate:** Pixels outside the frame are duplicated. This results in “smeared” edges but is useful when, for example, applying a blur because in that case black pixels would result in the unwanted blurring between the actual image and the black canvas.

### Clipping Mode

- **Domain:** Retains all pixels that might be moved out of the frame for later re-distorting.
- **Frame:** Pixels moved outside the frame are discarded.

### Output Distortion Map

Outputs the location of pixels as a warped screen-coordinate map.

### Camera Settings

The options known from the Camera 3D are duplicated here. They can either be set manually or connected to an already existing Camera 3D.

### Lens Distortion Model

Select the appropriate 3D Equalizer Lens Distortion model here: 3DE Classic Model, 3DE4 Anamorphic, 3DE4 Radial Fisheye, or 3DE4 Radial. Please consult the 3D Equalizer manual for further explanation. The sliders in the 3DE Classic LD Model are most likely best suited for manually applying (un)distortion, without having imported lens data.

### Supersampling [HiQ]

Sets the number of samples used to determine each destination pixel. As always, higher supersampling leads to higher render times. 1×1 bilinear is usually of sufficient quality, but with high lens distortion near the edges of the lens, there are noticeable differences to higher settings.

### Supersampling Mode [HiQ]

The type of sample done for each supersample. Nearest leads to a crisper but more aliased image. Bi-Linear gives a blurrier result.

### Load Distortion Data

Allows the user to load a Lens Distortion profile created, for example, by the 3D Equalizer.

#### How to Manually Determine Lens Distortion

In an ideal world, one would have exact lens parameters from each lens that was used during the shoot, and one could use those values to undistort the image. However, in the real world, those parameters have not been taken on set or don't match. Another approach is to use software like 3D Equalizer, which analyzes the footage and delivers a dataset that can be imported into the Lens Distort node right away.

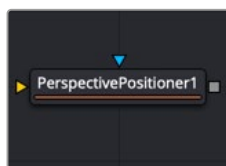
And finally, one could try to manually eyeball the amount of lens distortion using the control sliders. To do that, one could either look for horizontal or vertical lines in the footage that are supposed to be straight and straighten them out using the controls, or shoot a full-frame checkerboard pattern on set as a reference.

## Common Controls

### Settings Tab

The Settings tab in the Inspector is also duplicated in other Warp nodes. These common controls are described in detail at the end of this chapter in "The Common Controls" section.

# Perspective Positioner [PPn]



The Perspective Positioner node

## Perspective Positioner Node Introduction

The Perspective Positioner is the complementary node to the Corner Positioner node. It “unpins” an image by positioning corner points on a perspective distorted area, thereby removing the perspective from the image. This function can also be used to wobble and warp the image by animating the points over time.

### Inputs

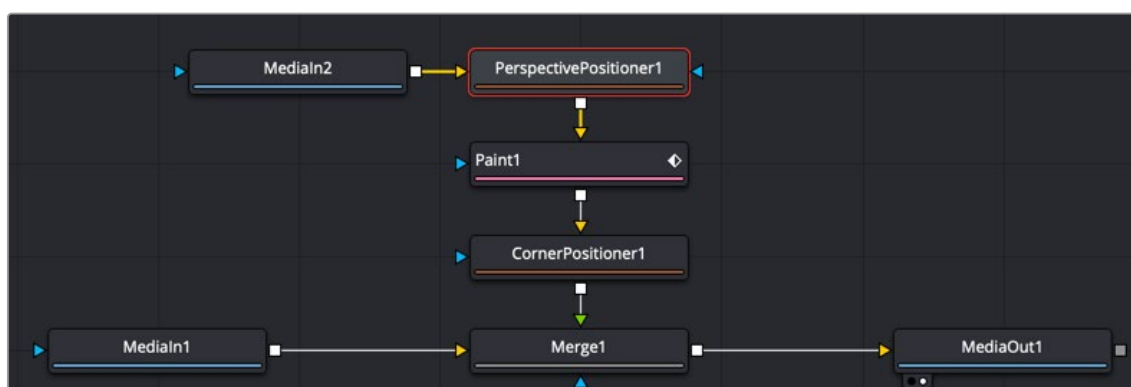
The two inputs on the Perspective Positioner node are used to connect a 2D image and an effect mask, which can be used to limit the transformed area.

**Input:** The orange input is used for the primary 2D image that is transformed.

**Effect Mask:** The blue input is for a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the transform to only those pixels within the mask. An effects mask is applied to the tool after the tool is processed.

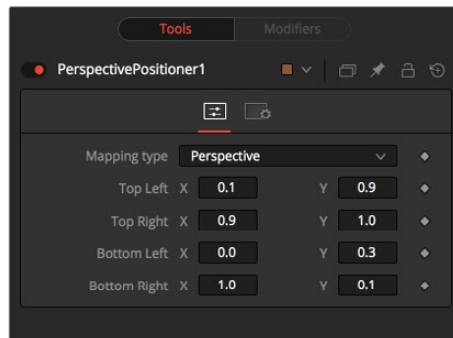
### Basic Node Setup

In the example below, the Perspective Positioner is used to unpin a perspective distorted area of the MediaIn2 in order to paint on the flat texture. The MediaIn2 is then corner pinned back into place. The Perspective Positioner and Corner Positioner do not concatenate, so some softness is introduced with these nodes.



The Perspective Positioner unpins an image to paint on a texture map.

## Inspector



The Perspective Positioner Controls tab

### Controls Tab

The Controls tab contains parameters for selecting vector channels and controlling how much distortion they apply to an image.

#### Mapping Type

The Mapping Type menu is used to select the type of transform used to distort the image. Bi-Linear is available for support of older projects. Leaving this on Perspective is strongly suggested since the Perspective setting maps the real world more accurately.

#### Corners X and Y

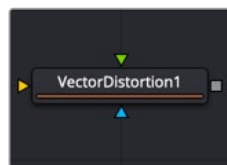
There are the four control points of the Perspective Positioner. Interactively drag these in the viewers to position each corner of the image. You can refine their position using the Top, Bottom, Left, and Right controls in the Inspector.

### Common Controls

#### Settings Tab

The Settings tab in the Inspector is also duplicated in other Warp nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

## Vector Distortion [Dst]



The Vector Distortion node

### Vector Distortion Node Introduction

The Vector Distortion node distorts the main source image along the X- and Y-axis separately, based on the vector channel data in the source image or vector channels from a second reference image.

## Inputs

There are three inputs on the Vector Distort node for the primary 2D image, the distort image with vector channels and an effect mask.

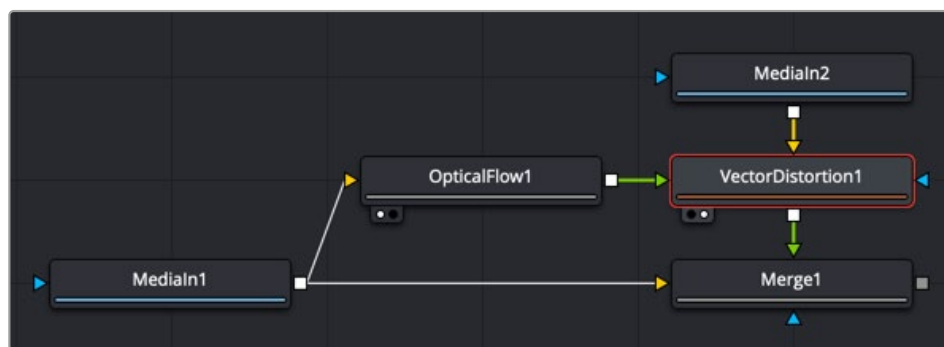
**Input:** The orange image input is a required connection for the primary image you wish to distort. If this image has vector channels, they are used in the distortion.

**Distort:** The green input is an optional distort image input used to distort the background image based on vector channels. Once connected, it overrides vector channels in the input image.

**Effect Mask:** The optional blue effect mask input expects a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the displacement to only those pixels within the mask. An effects mask is applied to the tool after it is processed.

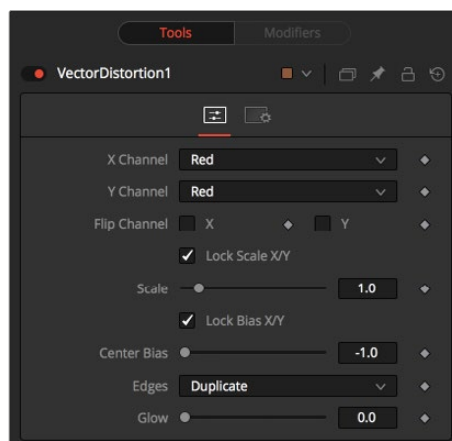
## Basic Node Setup

In the example below, the MediaIn1 node uses an Optical Flow node to generate the vector channels, which are then passed on to the Vector Distort node. MediaIn2 is distorted and composited over the background.



The Vector Distort node relies on the image having vector channels or an Optical Flow node to generate them.

## Inspector



The Vector Distortion Controls tab



## Controls Tab

The Controls tab contains parameters for selecting vector channels and controlling how much distortion they apply to an image.

### X Channel and Y Channel

These two menus are used to select which channel of the (green) distort image input is used to distort the X and Y channels. If no distort reference image is connected, then channels from the main orange input are used instead.

### Flip Channel X and Flip Channel Y

Use these checkboxes to flip the direction of the distortion along the specified axis.

### Lock Scale X/Y

Select this checkbox to separate the Scale slider into separate Scale X and Scale Y sliders.

### Scale

Use the Scale slider to apply a multiplier to the values of the distortion reference image.

### Lock Bias X/Y

Select this checkbox to separate the Bias slider into separate Bias X and Bias Y sliders.

### Center Bias

Use the Center Bias slider to shift or nudge the distortion along a given axis.

### Edges

This menu determines how the edges of the image are treated.

- **Canvas:** This causes the edges that are revealed by the shake to be the canvas color—usually transparent or black.
- **Duplicate:** This causes the edges to be duplicated, causing a slight smearing effect at the edges.

### Glow

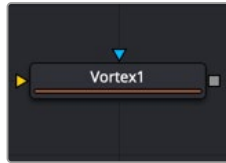
Use this slider to add a glow to the result of the vector distortion.

## Common Controls

### Settings Tab

The Settings tab in the Inspector is also duplicated in other Warp nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

# Vortex [Vtx]



The Vortex node

## Vortex Node Introduction

The Vortex effect appears as a swirling whirlpool in specified regions of the image. The Vortex can be made to move and grow by animating the various controls.

## Inputs

There are two inputs on the Vortex node for the primary 2D image and the effect mask.

**Input:** The orange image input is a required connection for the primary image you wish to swirl.

**Effect Mask:** The optional blue effect mask input expects a mask shape created by polylines, basic primitive shapes, paint strokes, or bitmaps from other tools. Connecting a mask to this input limits the swirling vortex to only those pixels within the mask. An effects mask is applied to the tool after it is processed.

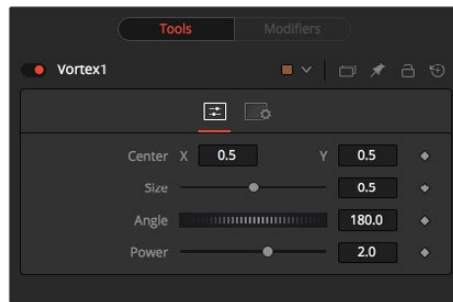
## Basic Node Setup

Below, the Vortex is applied to text for creating motion graphics. Since the Vortex will cause the text to swirl outside the text boundary, a Set Domain node is used to expand the text boundary, ensuring that the text is not cropped when the Vortex is applied.



The Vortex is used to create swirling whirlpool effects.

## Inspector



The Vector Distortion Controls tab

### Controls Tab

The Controls tab contains parameters for adjusting the position, size, and strength of the Vortex effect.

#### Center X and Y

This control is used to position the center of the Vortex effect on the image. The default is 0.5, 0.5, which positions the effect in the center of the image.

#### Size

Size changes the area affected by the Vortex. You can drag the circumference of the effect in the viewer or use the Size slider.

#### Angle

Drag the rotation handle in the viewer or use the thumbwheel control to change the amount of rotation in the Vortex. The higher the angle value, the greater the swirling effect.

#### Power

Increasing the Power slider makes the Vortex smaller but tighter. It effectively concentrates it inside the given image area.

### Common Controls

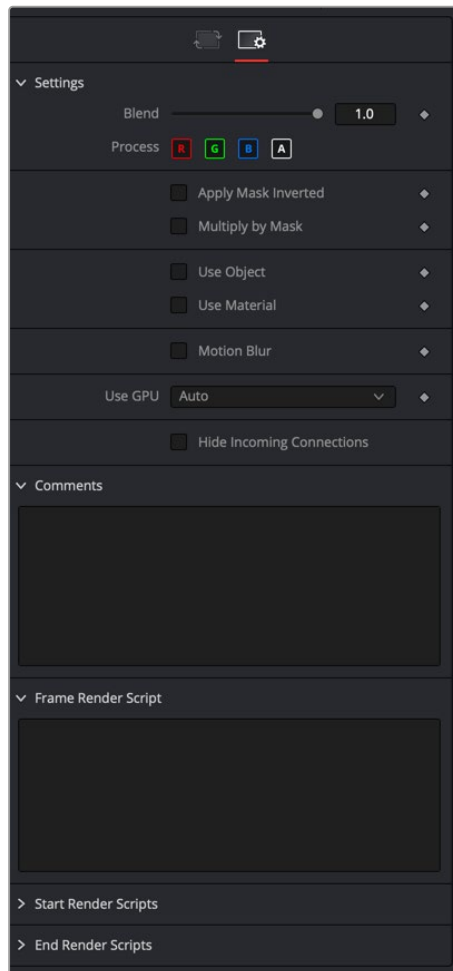
#### Settings Tab

The Settings tab in the Inspector is also duplicated in other Warp nodes. These common controls are described in detail in the following “The Common Controls” section.

# The Common Controls

Nodes that handle Warp operations share a number of identical controls in the Inspector. This section describes controls that are common among Warp nodes.

## Inspector



Warp Common Controls tab

## Settings Tab

The Settings tab in the Inspector can be found on every tool in the Warp category. The Settings controls are even found on third-party Warp-type plugin tools. The controls are consistent and work the same way for each tool.

### Blend

The Blend control is used to blend between the tool's original image input and the tool's final modified output image. When the blend value is 0.0, the outgoing image is identical to the incoming image. Normally, this will cause the tool to skip processing entirely, copying the input straight to the output.

### Process When Blend Is 0.0

The tool is processed even when the input value is zero. This can be useful if the node is scripted to trigger a task, but the node's value is set to 0.0.

### Red/Green/Blue/Alpha Channel Selector

These four buttons are used to limit the effect of the tool to specified color channels. This filter is often applied after the tool has been processed.

For example, if the Red button on a Blur tool is deselected, the blur will first be applied to the image, and then the red channel from the original input will be copied back over the red channel of the result.

There are some exceptions, such as tools for which deselecting these channels causes the tool to skip processing that channel entirely. Tools that do this will generally possess a set of identical RGBA buttons on the Controls tab in the tool. In this case, the buttons in the Settings and the Controls tabs are identical.

### Apply Mask Inverted

Enabling the Apply Mask Inverted option inverts the complete mask channel for the tool. The mask channel is the combined result of all masks connected to or generated in a node.

### Multiply by Mask

Selecting this option will cause the RGB values of the masked image to be multiplied by the mask channel's values. This will cause all pixels of the image not included in the mask (i.e., set to 0) to become black/transparent.

### Use Object/Use Material (Checkboxes)

Some 3D software can render to file formats that support additional channels. Notably, the EXR file format supports Object ID and Material ID channels, which can be used as a mask for the effect. These checkboxes determine whether the channels will be used, if present. The specific Material ID or Object ID affected is chosen using the Object and Material ID sliders explained below.

### Correct Edges

This checkbox appears only when the Use Object or Use Material checkboxes are selected. It toggles the method used to deal with overlapping edges of objects in a multi-object image. When enabled, the Coverage and Background Color channels are used to separate and improve the effect around the edge of the object. If this option is disabled (or no Coverage or Background Color channels are available), aliasing may occur on the edge of the mask.

For more information on the Coverage and Background Color channels, see *Chapter 78, "Understanding Image Channels,"* in the DaVinci Resolve Reference Manual, or Chapter 18 in the Fusion Reference Manual.

### Object ID/Material ID (Sliders)

Use these sliders to select which ID will be used to create a mask from the object or material channels of an image. Use the Sample button in the same way as the Color Picker: to grab IDs from the image displayed in the viewer. The image or sequence must have been rendered from a 3D software package with those channels included.

## Motion Blur

- **Motion Blur:** This toggles the rendering of Motion Blur on the tool. When this control is toggled on, the tool's predicted motion is used to produce the motion blur caused by the virtual camera's shutter. When the control is toggled off, no motion blur is created.
- **Quality:** Quality determines the number of samples used to create the blur. A quality setting of 2 will cause Fusion to create two samples to either side of an object's actual motion. Larger values produce smoother results but increase the render time.
- **Shutter Angle:** Shutter Angle controls the angle of the virtual shutter used to produce the motion blur effect. Larger angles create more blur but increase the render times. A value of 360 is the equivalent of having the shutter open for one full frame exposure. Higher values are possible and can be used to create interesting effects.
- **Center Bias:** Center Bias modifies the position of the center of the motion blur. This allows for the creation of motion trail effects.
- **Sample Spread:** Adjusting this control modifies the weighting given to each sample. This affects the brightness of the samples.

## Use GPU

The Use GPU menu has three settings. Setting the menu to Disable turns off GPU hardware-accelerated rendering. Enabled uses the GPU hardware for rendering the node. Auto uses a capable GPU if one is available and falls back to software rendering when a capable GPU is not available.

## Hide Incoming Connections

Enabling this checkbox can hide connection lines from incoming nodes, making a node tree appear cleaner and easier to read. When enabled, empty fields for each input on a node will be displayed in the Inspector. Dragging a connected node from the node tree into the field will hide that incoming connection line as long as the node is not selected in the node tree. When the node is selected in the node tree, the line will reappear.

## Comments

The Comments field is used to add notes to a tool. Click in the empty field and type the text. When a note is added to a tool, a small red square appears in the lower-left corner of the node when the full tile is displayed, or a small text bubble icon appears on the right when nodes are collapsed. To see the note in the Node Editor, hold the mouse pointer over the node to display the tooltip.

## Scripts

Three Scripting fields are available on every tool in Fusion from the Settings tab. They each contain edit boxes used to add scripts that process when the tool is rendering. For more details on scripting nodes, please consult the Fusion scripting documentation.